SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	**Assessment:**	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	**Total Marks:**	100
CO5 Statement:	*Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.*		
Student Name:	Jeeva Raghul .M	**Reg. No.:**	192524487
Section / Batch:	B Tech AIDS	**Date:**	28/08/2026

Code of Conduct

I ____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____________________

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
		GRAND TOTAL:	**100 Marks**

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QUESTIONS

Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST & Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q1 **Minimum Spanning Tree - Kruskal's Algorithm** 20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

a. All zones must be connected

b. Total installation cost must be minimized

c. No redundant connections (no cycles) allowed

d. The network must remain fully connected

Your Answer

```
    MakeSet(v)
For each edge (u, v) in E
    If FindSet(u) ≠ FindSet(v)
        MST ← MST ∪ {(u, v)}
        Union(u, v)
Return MST
```

Save Answer   Next →   ✓ Saved

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QUESTIONS

Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST & Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

Q2 **Graph - Topological Sort** 20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.
Constraints:
a. Detect cycle before sorting
b. Use DFS approach

Your Answer

```
    MakeSet(v)
For each edge (u, v) in E
    If FindSet(u) ≠ FindSet(v)
        MST ← MST ∪ {(u, v)}
        Union(u, v)
Return MST
```

Save Answer   Next

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QUESTIONS

Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST & Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q3 **Shortest Path Algorithm - MST & Dijkstra's Algorithm** 20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.
Constraints:
a. Use MST for infrastructure
b. Use shortest path for routing

Your Answer

```
For each vertex v in V
    If visited[v] = false
        DFS(v)

Return order
```

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QUESTIONS

Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST & Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

Q4 **Shortest Path Algorithm - Dijkstra's Algorithm** 20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

```
  For each neighbor v of u in MST
      If dist[v] > dist[u] + weight(u, v)
          dist[v] ← dist[u] + weight(u, v)

Return MST, dist
```

Save Answer   Next →

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QUESTIONS

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20 marks

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Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST & Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

Q5 **Minimum Spanning Tree - Prim's or Kruskal's Algorithm** 20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST. (20 Marks)
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

```
Algorithm SecondBestMST(G)
Input: Graph G(V, E) with weights
Output: Second Best Minimum Spanning Tree

MST ← KruskalMST(G)
bestWeight ← weight(MST)
```

Save Answer